

FONCTIONS D'UNE VARIABLE RÉELLE

Bibliothèque matplotlib

Correction Exercices

Exercice 1 :

```
from pylab import*
xlim(0,10)
ylim(0,10)
x=[2,5,8,2]
y=[2,8,2,2]
plot(x,y,linewidth=3,*'r')
show()
```

Exercice 2 :

```
from pylab import*
xlim(0,10)
ylim(0,10)
x=[3,7,7,3,3]
y=[3,3,7,7,3]
plot(x,y,linewidth=3,*'r')
show()
```

Exercice 3 :

```
from pylab import*
xlim(0,20)
ylim(2,4)
y=3
for i in range(10):
    x=5+i
    plot(x,y,'ob')
show()
```

Exercice 4 :

```
from pylab import*
xlim(0,20)
ylim(2,6,0.5)
y=3
for i in range(10):
    x=5+i
    plot(x,y,'ob')
y=5
for i in range(10):
    x=5+i
    plot(x,y,'ob')
show()
```

Exercice 5 :

```
from pylab import*
xlim(0,20,5)
ylim(2,30,5)
for j in range(10):
    y=10+j
    for i in range(10):
        x=5+i
        plot(x,y,'ob')
show()
```

Exercise 6 :

```
from pylab import*
xlim(0,50,10)
ylim(2,50,10)
for i in range (31):
    x=10+i
    y=x
    plot(x,y,'ob')
show()
```

Exercise 7 :

```
from pylab import*
xlim(0,25,5)
ylim(0,45,5)
for i in range (21):
    x=i
    y_1=2*x
    y_2=x
    y_3=0.5*x
    plot(x,y_1,'dr')
    plot(x,y_2,'ob')
    plot(x,y_3,'*r')
show()
```

Exercise 8 :

1.

```
from pylab import*
xlim(-3,3,1)
ylim(-30,30,5)
liste_x=[]
liste_y=[]
for i in range (100):
    x=-3+0.06*i
    y=x**3
    liste_x.append(x)
    liste_y.append(y)
plot(liste_x,liste_y,'k')
show()
```

2.

```
from pylab import*
xlim(-5,5,1)
ylim(-50,50,1)
liste_x=[]
liste_y=[]
for i in range(250):
    x=-5+0.02*i
    y=1/x
    liste_x.append(x)
    liste_y.append(y)

for i in range(250):
    x=0.02+0.02*i
    y=1/x
    liste_x.append(x)
    liste_y.append(y)
plot(liste_x,liste_y,'k')
show()
```